

Local Planning and Evaluation of Evasive Maneuvers

Project Presentation

Presented by

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Introduction & Motivation

Background:

Autonomous vehicles operate in highly dynamic traffic environments where unexpected events such as stationary obstacles, slow-moving vehicles, and pedestrians require immediate and safe decision-making. The vehicle must generate a new trajectory within milliseconds while ensuring passenger comfort and complying with vehicle dynamic constraints.

Key Challenges:

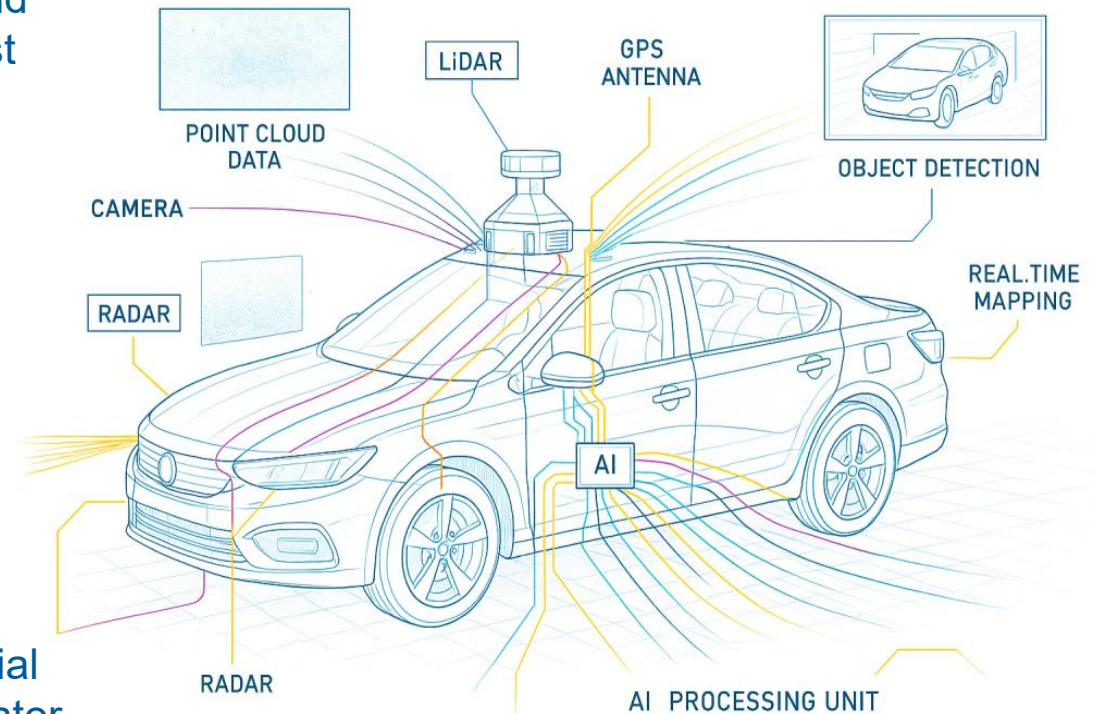
1. Static obstacles
2. Dynamic obstacles
3. Pedestrian interaction
4. Real-time trajectory generation
5. Vehicle dynamic constraints

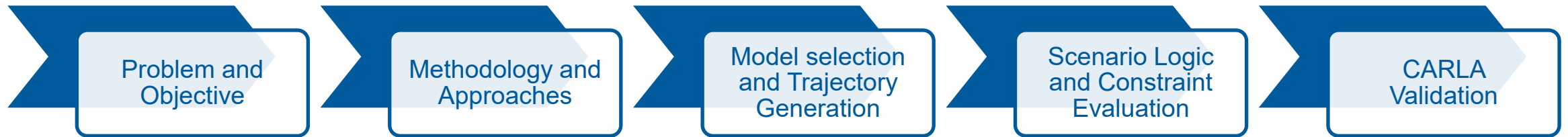
Project Goal:

Develop a real-time, collision-free trajectory planner using quintic polynomial interpolation and a kinematic bicycle model, validated in the CARLA simulator.

Research Focus:

Generate safe, smooth, collision-free trajectories in real time while satisfying vehicle dynamic and safety constraints.

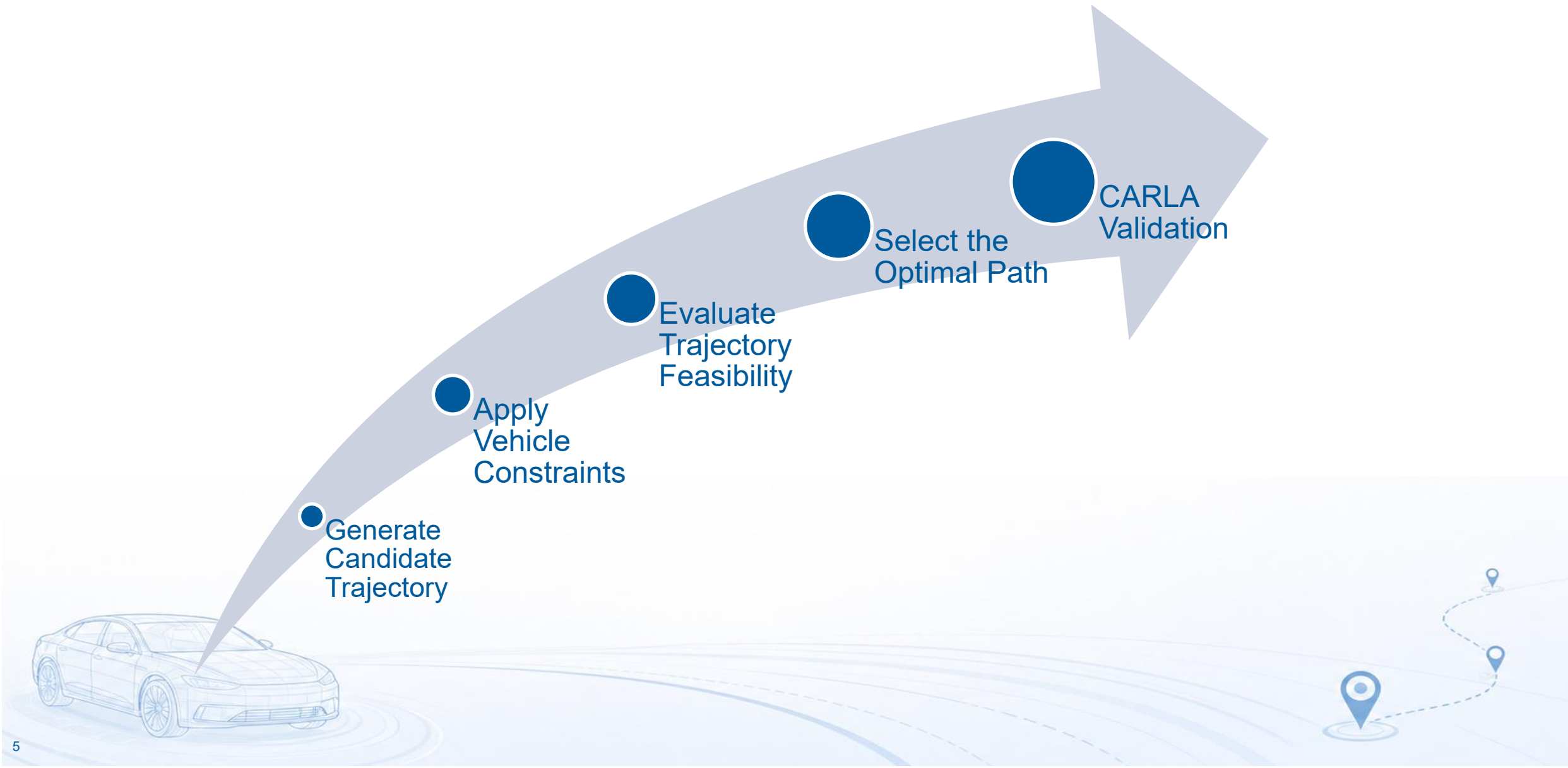




Objectives & Challenges



Scope of Work



Types of Method



Sampling-Based Methods

RRT / RRT*
Hybrid A*
Flexible search

Advantages

- Handles complex environments
- Obstacle avoidance

Limitations

- Non-smooth paths
- High computation
- Runtime issues

Curve Interpolation Methods

Quintic Polynomial
Sinusoidal Curves
Smooth paths

Advantages

- Smooth trajectories
- Low computation
- Easy constraint handling

Limitations

- Less flexible in complex scenes

Machine Learning / Deep Learning

Neural Networks
LSTM Models
Learns from data

Advantages

- Captures complex driving behavior
- Adaptive

Limitations

- Large dataset required
- Limited interpretability
- No hard safety guarantees



Constraint-Based Optimal Trajectory Planning

- Generates smooth and dynamically feasible trajectories
- Ensures collision-free obstacle avoidance.
- Incorporates vehicle dynamic and safety constraints
- Optimizes trajectory using a multi-objective cost function
- Suitable for real-time autonomous driving applications



Kinematic Bicycle Model

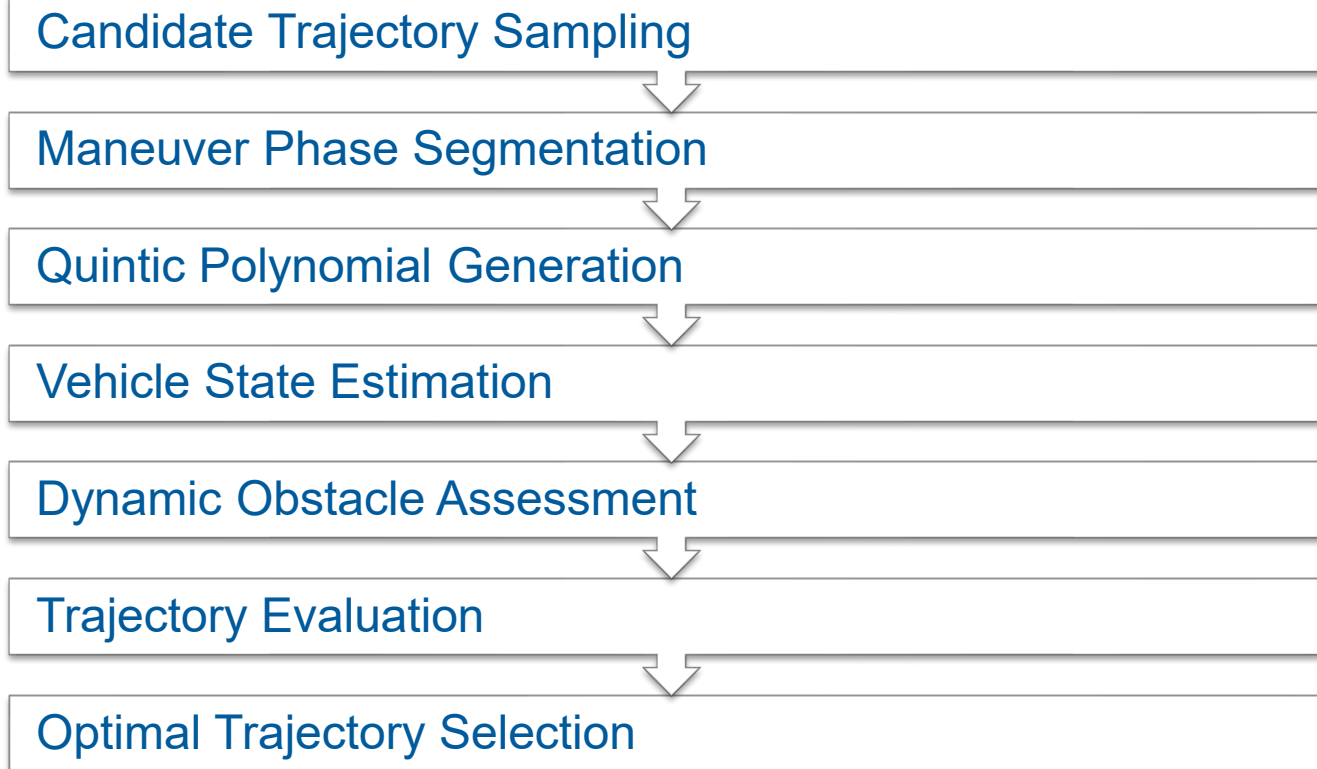
- Simplifies vehicle motion by representing the four-wheel vehicle as a two-wheel equivalent model.
- Provides an accurate approximation for **low- and medium-speed** trajectory planning.
- Enables efficient computation of vehicle states with **low computational**
- Well suited for **real-time autonomous lane-change and obstacle avoidance** applications.

How it is used

- $\psi = \tan^{-1}(y')$
- $\kappa = \frac{y''}{(1+(y')^2)^{3/2}}$
- $\delta = \tan^{-1}(L\kappa)$
- $a_y = v_{ego}^2 \kappa$



Trajectory Generation



Constraint Handling



Lateral acceleration
limit

Steering angle limit

Minimum obstacle
clearance

Relative speed
condition

Pedestrian safety
clearance



Mathematical Approach



1. Quintic polynomial

$$y(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5$$

2. Phase points

$$x_1 = \frac{x_{obs} - \frac{L_{obs} + L_{ego}}{2} - d_{start}}{1 - \frac{v_{obs}}{v_{ego}}}$$

$$x_4 = \frac{x_{obs} + \frac{L_{obs} + L_{ego}}{2} + d_{return}}{1 - \frac{v_{obs}}{v_{ego}}}$$

3. Bicycle model

$$\begin{aligned}\psi &= \tan^{-1}(y') \\ \kappa &= \frac{y''}{(1 + (y')^2)^{3/2}} \\ \delta &= \tan^{-1}(L\kappa) \\ a_y &= v_{ego}^2 \kappa\end{aligned}$$



4. Constraints

$$J = w_1 a_y^2 + w_2 L + \frac{w_3}{d_{obs}} + \frac{w_4}{d_{ped}} - w_5 v$$

5. Constraint-Based Trajectory Validation

Feasible trajectory = {True: $L_{pass} > 0$, $a_{y,max} \leq a_{y,lim}$, $\delta_{max} \leq \delta_{lim}$, $d_{clear} \geq d_{min}$
False: otherwise

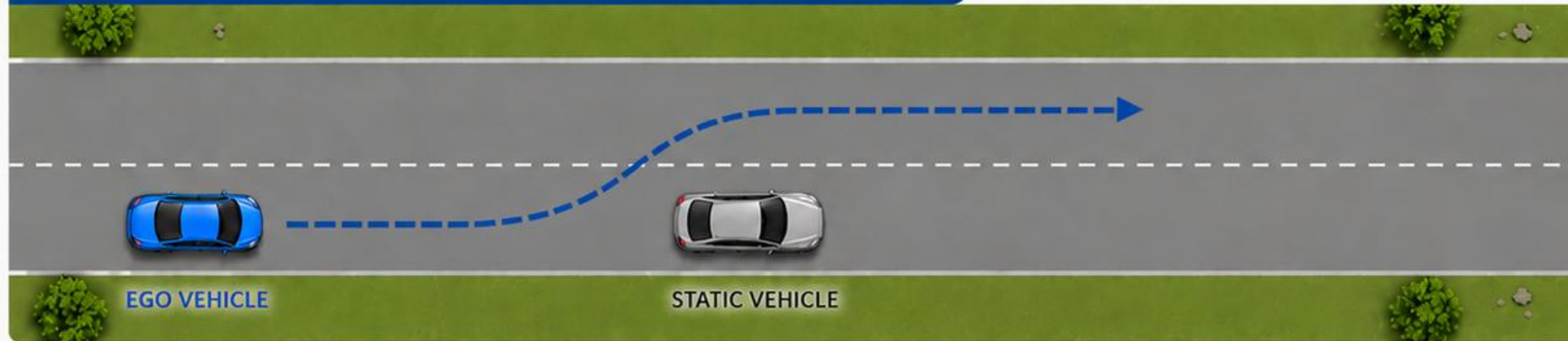


SCENARIO 1

Static Obstacle - Lane Change



SCENARIO 1: STATIC OBSTACLE – LANE CHANGE



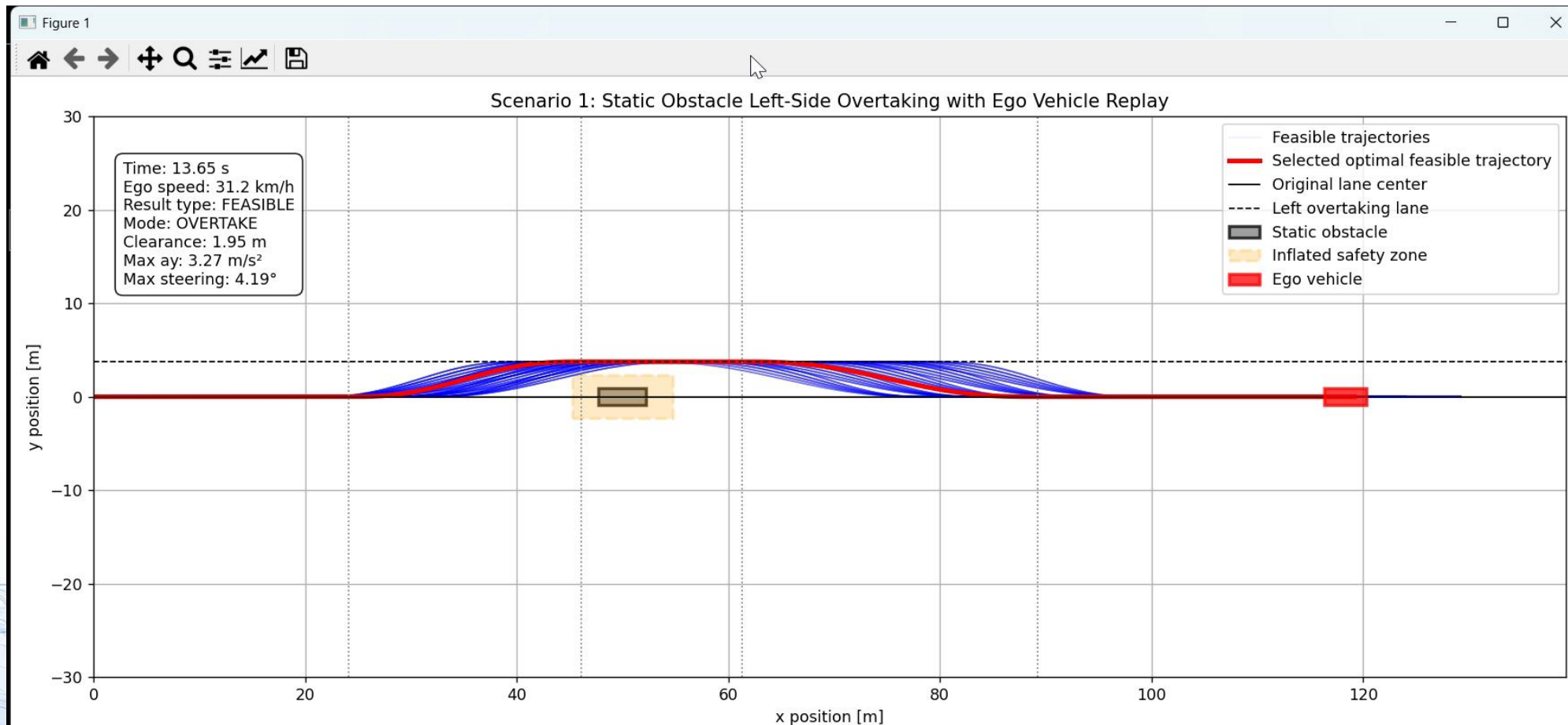
Legend

-  Ego Vehicle
-  Static Vehicle
-  Planned Trajectory

Result Analysis

Scenario 1

- Implemented kinematic bicycle
- Smooth lane-change trajectory using polynomial path shaping.
- Obstacle vehicle at $x = 50$ m in original lane.
- Target lane center set to $y = 3.75$ m.
- Generated feasible trajectories for evaluation.
- All feasible trajectories analyzed based on safety and comfort constraints.

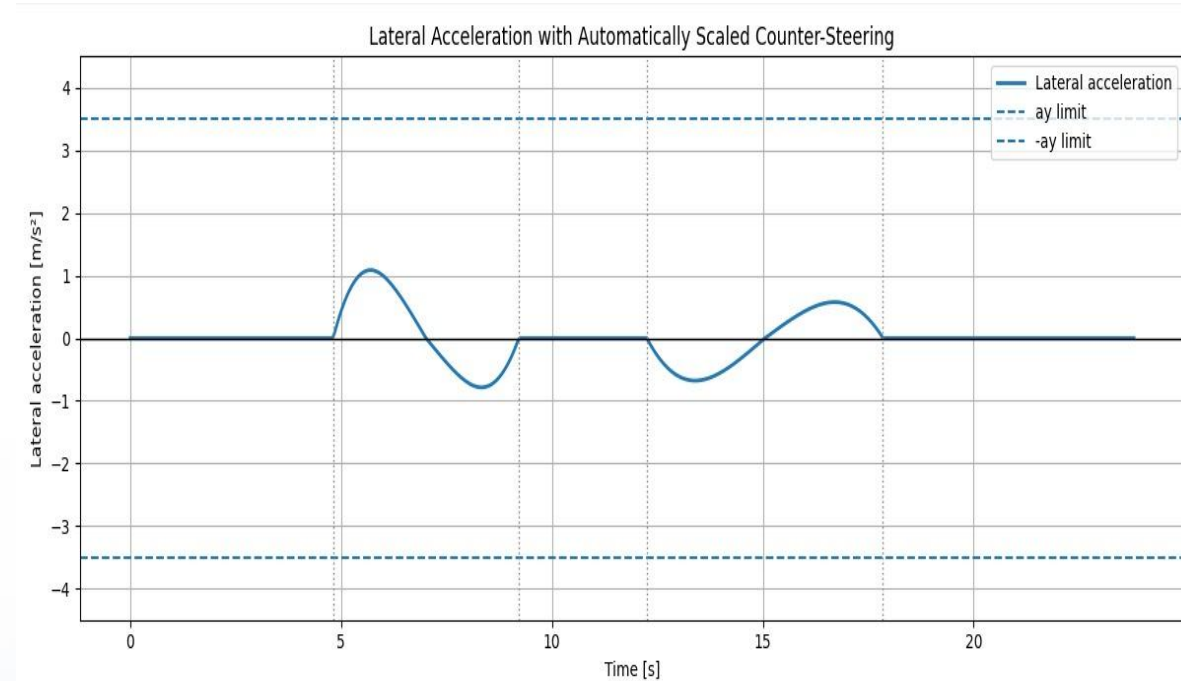
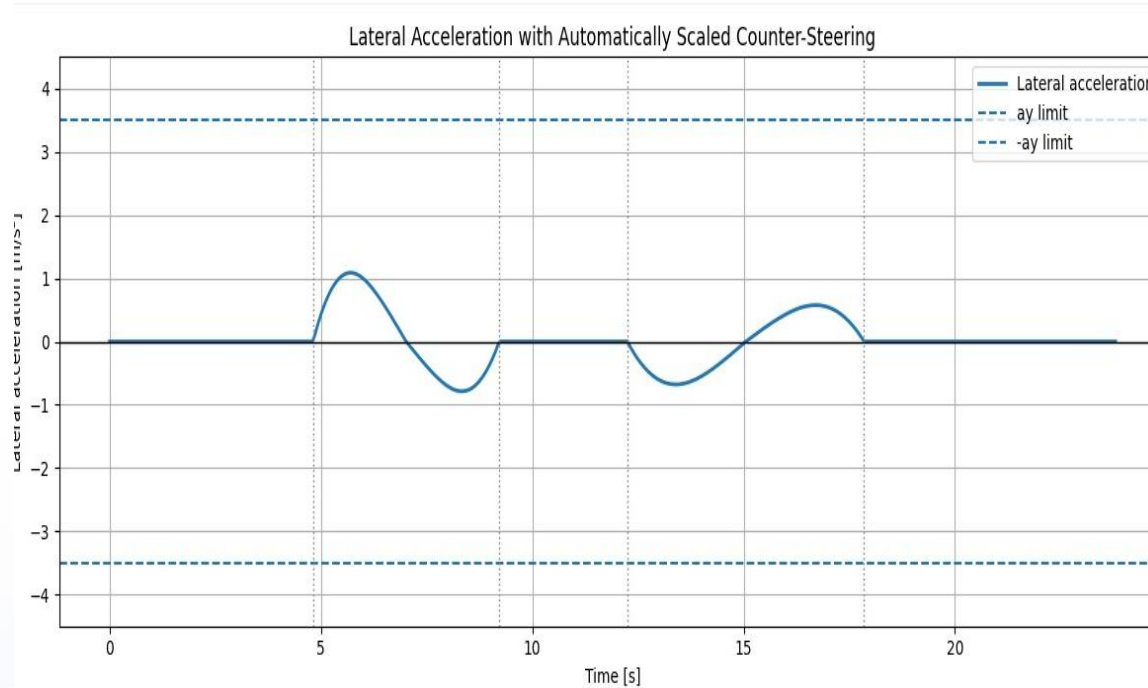


Scenario 1



- Added automatic overtaking decision logic.
- Introduced minimum safety gap between ego and obstacle vehicle.
- Safety gap range maintained between 15 m – 25 m.

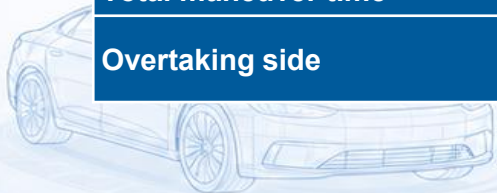
- Improved steering angle smoothness for better stability.
- Added clearance checking to avoid collision risk.
- Optimized trajectory based on cost function and comfort.



Scenario 1

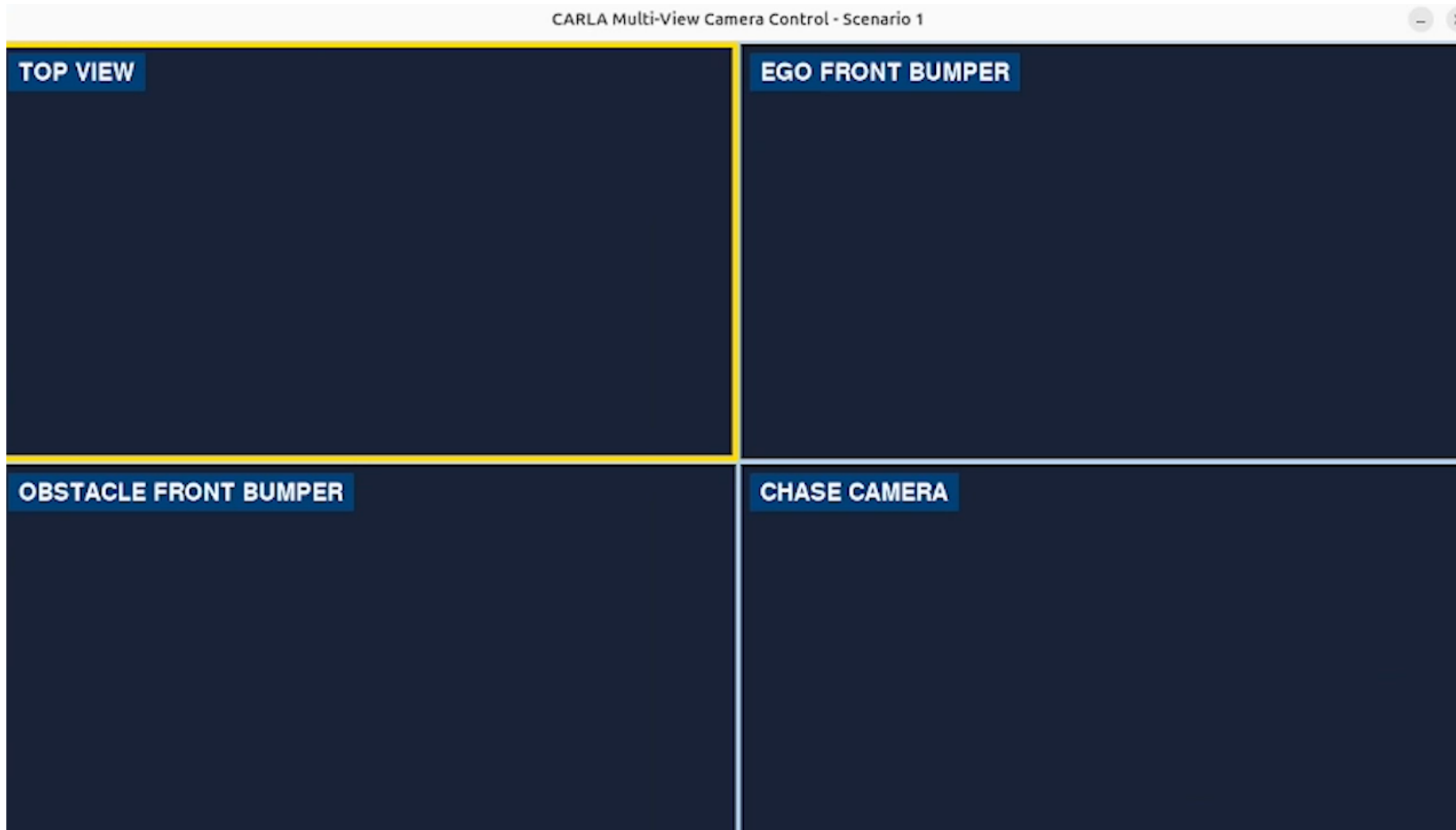


Metric	Value	Limit / Target	Status
Feasible trajectories	2704	—	—
Ego speed	18.0 km/h	—	—
Outward lane-change length	22.0 m	—	—
Return lane-change length	28.0 m	—	—
Passing length	15.17 m	—	—
Cost value	11.9037	—	—
Max lateral acceleration	1.0876 m/s ²	—	—
Maximum steering angle	4.1952°	—	—
Outward main ay	1.0876 m/s ²	≤ 3.5 m/s ²	✓ PASS
Outward counter ay	0.7856 m/s ²	< main ay	✓ PASS
Return main ay	0.6783 m/s ²	≤ 3.5 m/s ²	✓ PASS
Return counter ay	0.5765 m/s ²	< main ay	✓ PASS
Outward counter scale α	0.722	[0.40, 0.85]	✓ PASS
Return counter scale α	0.850	[0.40, 0.85]	✓ PASS
Lane-change time	~4.4 s	—	—
Passing time	3.033 s	—	—
Return time	5.6 s	—	—
Total maneuver time	13.033 s	—	—
Overtaking side	LEFT (positive y)	LEFT only	✓ PASS



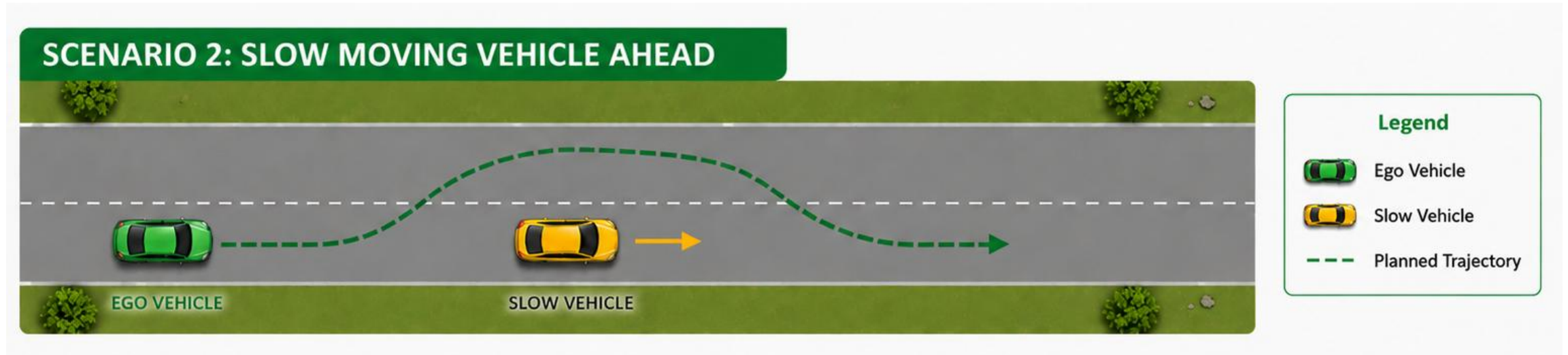
CARLA VALIDATION

Scenario 1



SCENARIO 2

Slow-Moving Vehicle Ahead

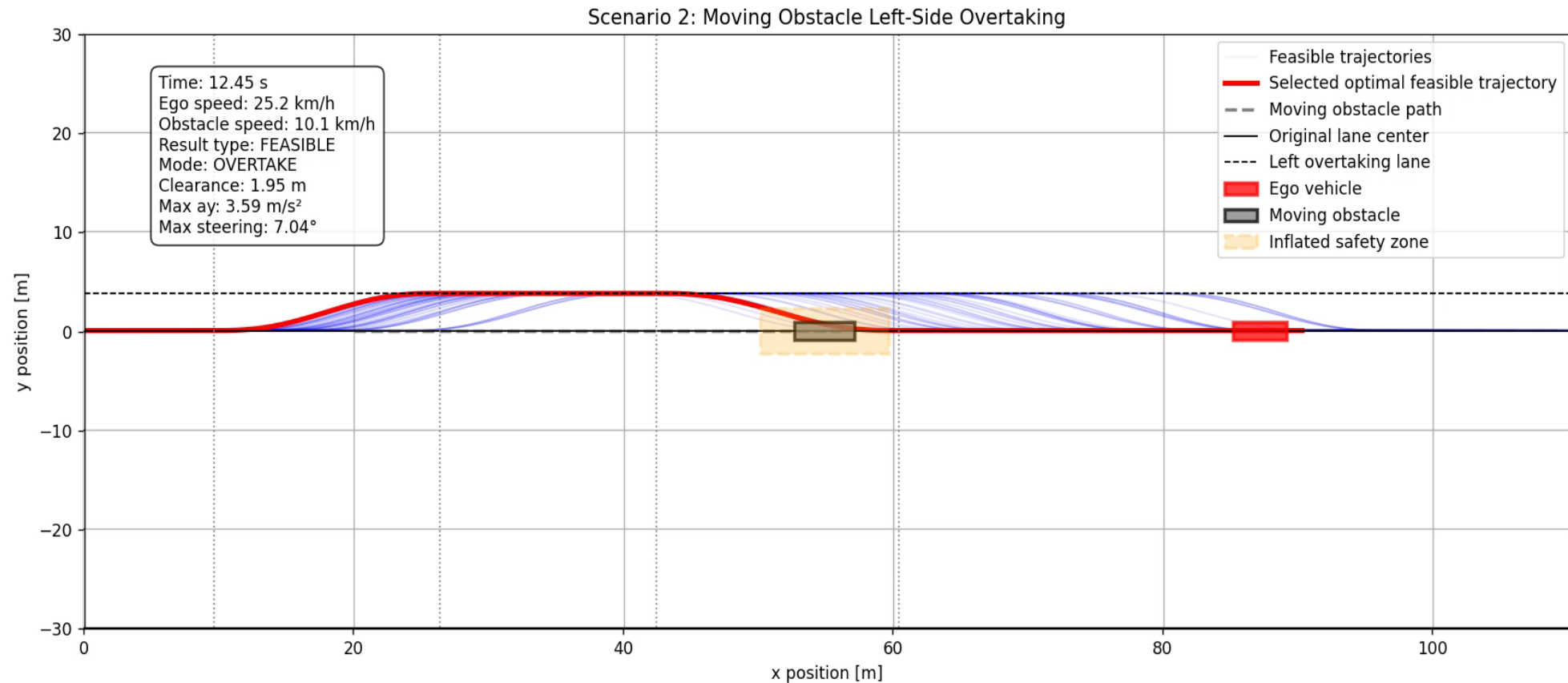


Result Analysis

Scenario 2



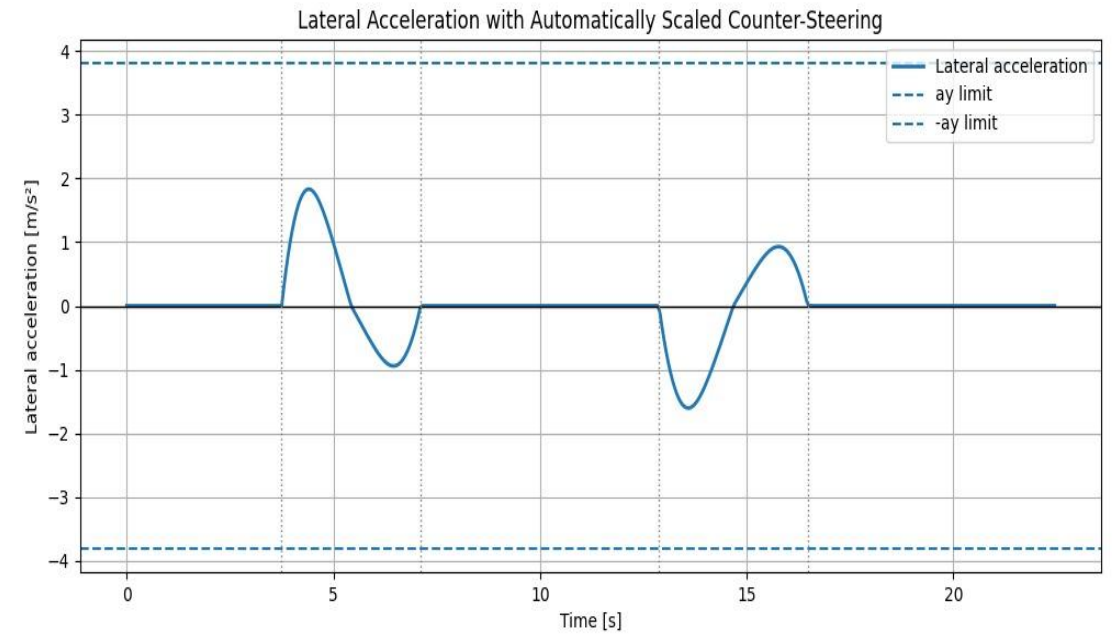
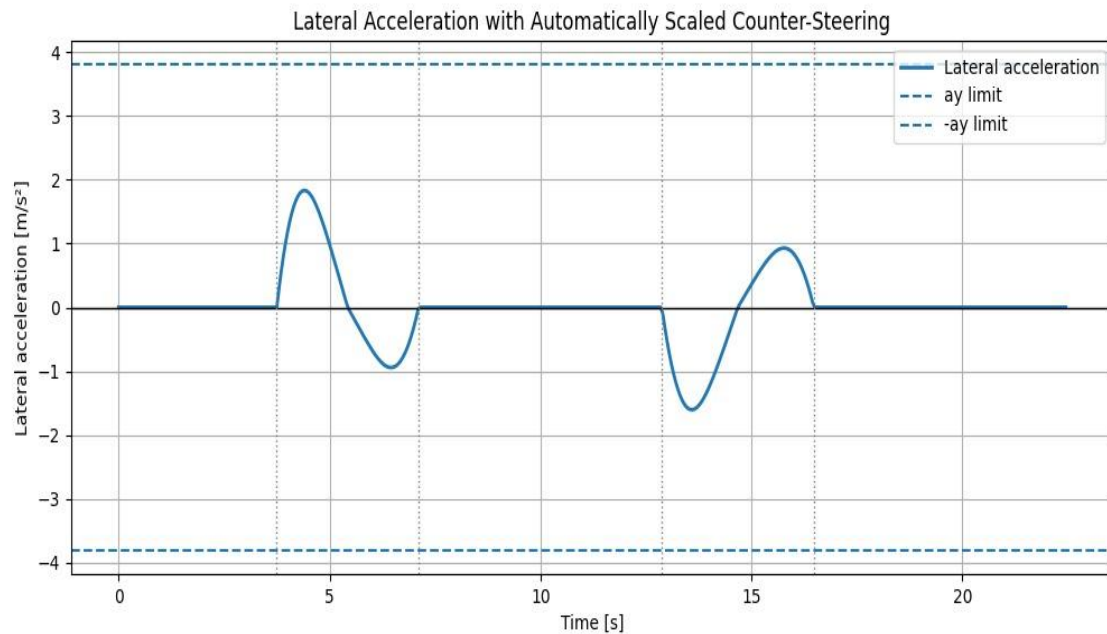
- System now can safely choose braking if overtaking is not feasible
- Added braking trajectory evaluation
- Computes required deceleration in unsafe conditions
- Checks comfort braking feasibility (max 4 m/s^2)
- Ensures system never forces unsafe overtaking
- Improved Vehicle Stability



Scenario 2

- Relative-speed-based overtaking of a moving obstacle.
- Quintic polynomial trajectory generation for smooth lane transitions.
- Dynamic obstacle position prediction during the maneuver.
- Real-time clearance monitoring throughout overtaking.

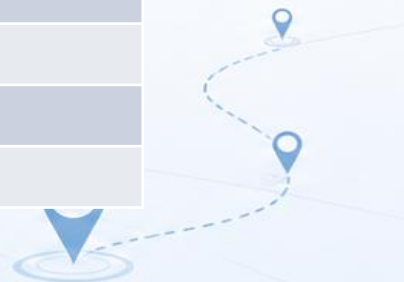
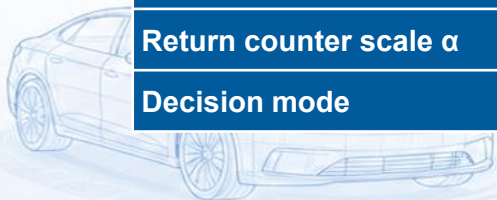
- Multi-constraint validation (clearance, steering angle, lateral acceleration, passing length).
- Cost-function-based optimal trajectory selection.
- Adaptive counter-steering for improved vehicle stability.
- Automatic braking fallback when overtaking is unsafe



Scenario 2

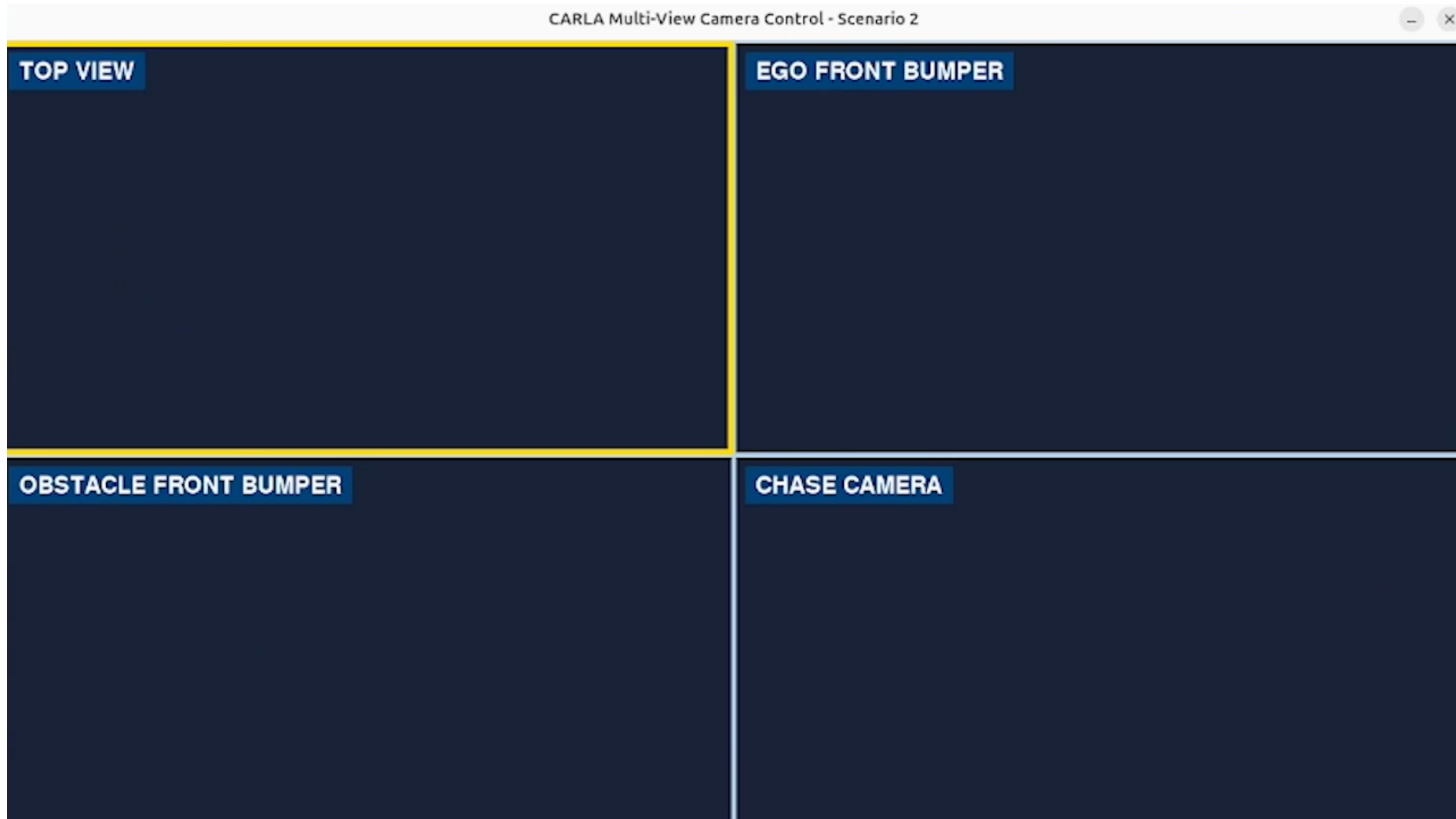


Metric	Value	Limit / Target	Status
Ego speed	18.0 km/h (5.0 m/s)	4.0–5.0 m/s	✓
Feasible trajectories	2592	—	—
Obstacle speed	10.1 km/h (2.8 m/s)	2.8–3.8 m/s	✓
Relative speed	2.2 m/s	≥ 1.0 m/s	✓
Outward lane-change length	16.8 m	—	—
Return lane-change length	18.0 m		
Passing length	28.84 m	—	—
Cost value	15.2094	—	—
Max lateral acceleration	1.8329 m/s ²	≤ 3.8 m/s ²	✓ PASS
Max steering angle	7.047°	≤ 25°	✓ PASS
Minimum clearance	1.95 m	≥ 0.5 m	✓ PASS
Outward main a_y	1.8329 m/s ²	—	—
Return main a_y	1.6045 m/s ²	—	—
Outward counter scale α	0.515	[0.40, 0.85]	✓
Return counter scale α	0.579	[0.40, 0.85]	✓
Decision mode	OVERTAKE	—	✓



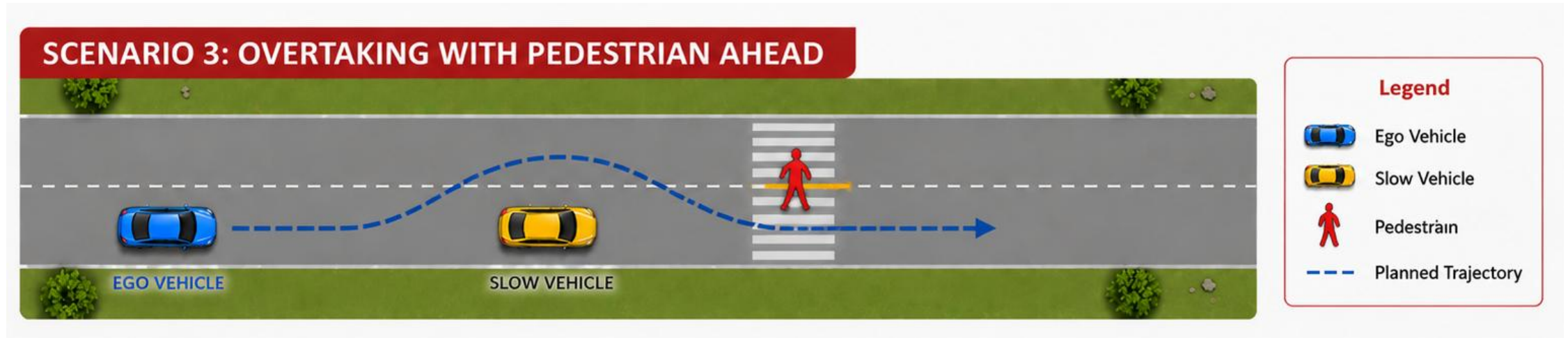
CARLA VALIDATION

Scenario 2



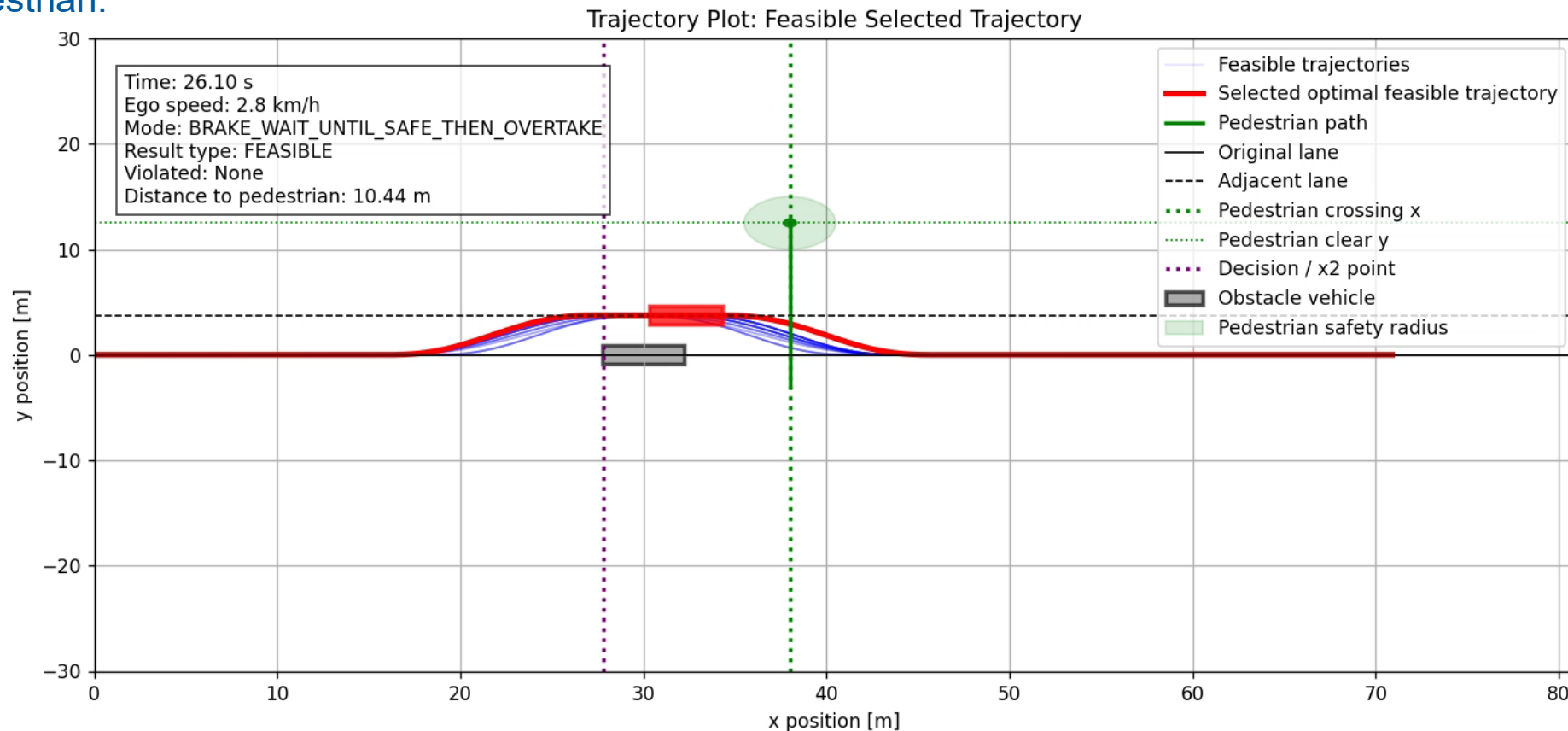
SCENARIO 3

Overtaking with Pedestrian Crossing



Scenario 3: Simulation

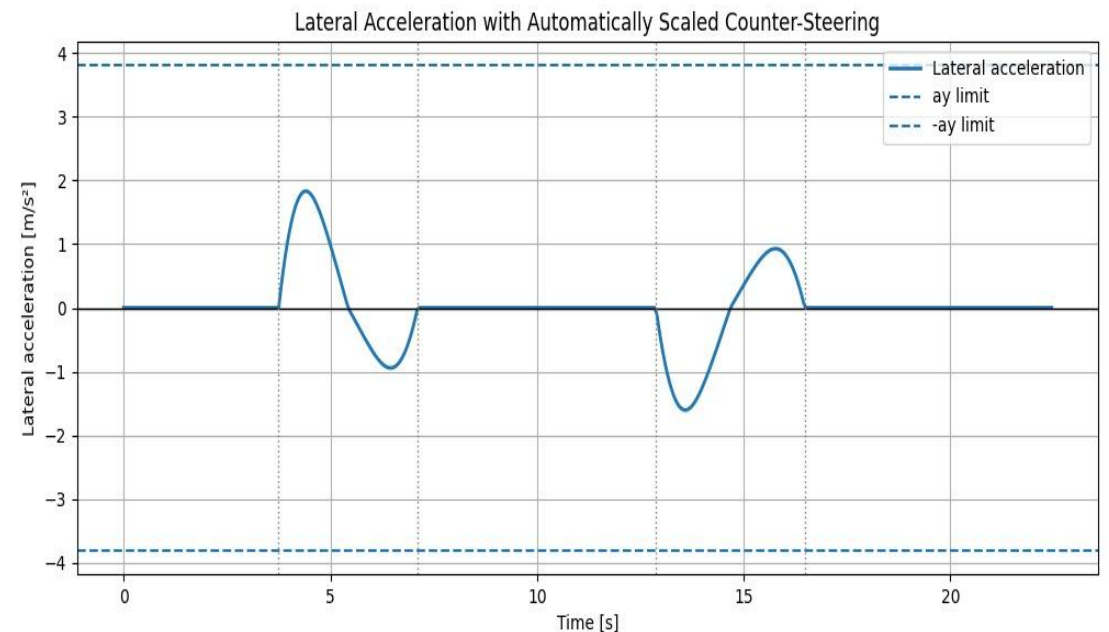
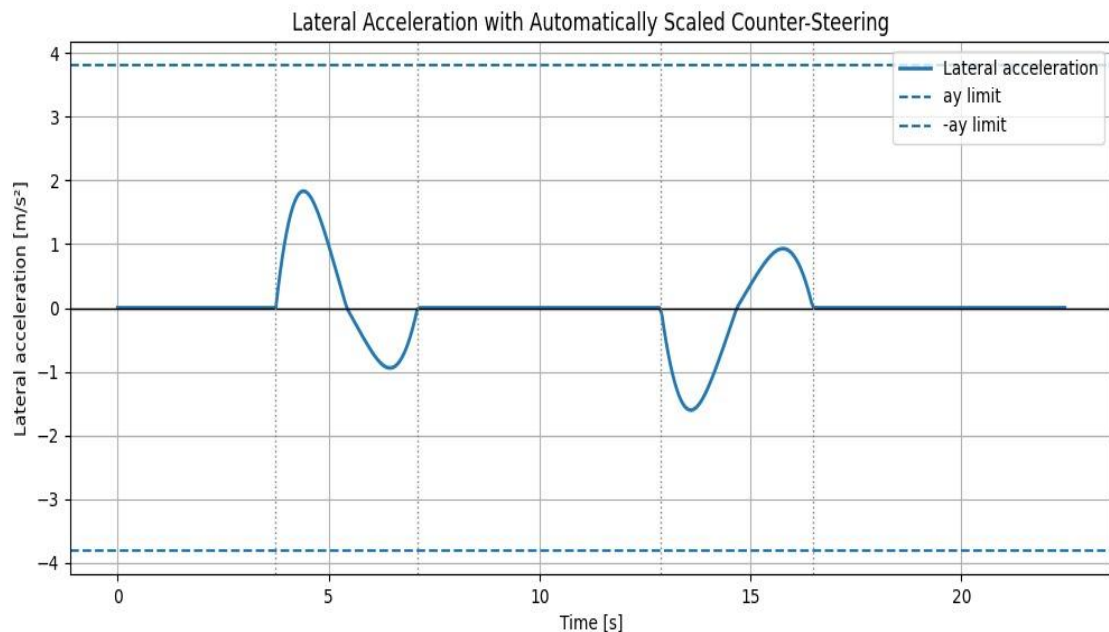
- Implemented automatic **Wait–Then–Overtake** decision strategy.
- Predicts pedestrian movement before lane-change execution.
- Generates collision-free trajectories around static obstacle and pedestrian.
- Checks pedestrian safety zone before overtaking.
- Validates all safety constraints before maneuver execution.
- Ensures safe and comfortable trajectory with no constraint violations



Scenario 3



- Integrated pedestrian crossing prediction into the local trajectory planner.
- Implemented Wait / Continue decision logic based on pedestrian movement.
- Evaluated future pedestrian position before executing the overtaking maneuver.
- Combined static obstacle avoidance with pedestrian safety assessment.
- Enforced a pedestrian safety radius to prevent conflicts during crossing.
- Performed simultaneous validation of obstacle clearance, steering angle, and lateral acceleration.
- Selected the optimal trajectory using a multi-objective cost function considering safety and driving comfort.
- Validated the complete scenario in CARLA with synchronized vehicle and pedestrian motion.



Scenario 3

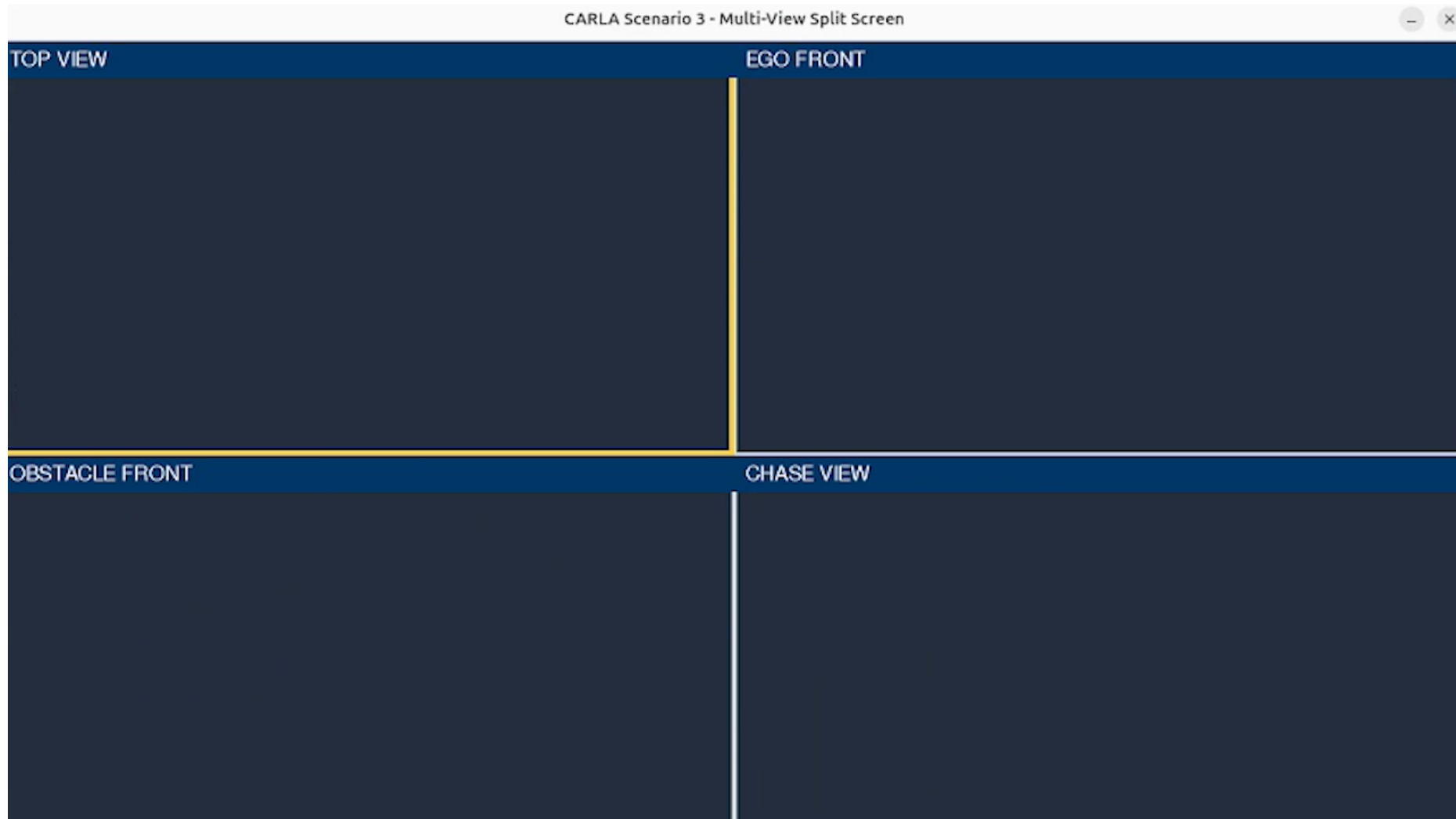


Metric	Value	Limit / Target	Status
Total trajectories	1296	—	—
Feasible trajectories	56	—	—
Rejected bypass length	1,232	—	Most rejections
Rejected by lateral accel.	0	—	—
Rejected by obstacle clearance	1152	—	—
Ego speed	4.0 m/s (14.4 km/h)	4.0–5.5 m/s	✓
Lane-change out length	12.000 m	—	—
Lane-change return length	12.000 m	—	—
Total maneuver length	28.500 m	—	—
Cost	21.869	Minimum	✓
Max lateral acceleration	2.221 m/s ²	≤ 3.8 m/s ²	✓ PASS
Max steering angle	13.171°	≤ 25°	✓ PASS
Obstacle clearance	1.779 m	≥ 0.5 m	✓ PASS
Future pedestrian clearance	3.543 m	≥ safety threshold	✓ PASS
Ped margin at decision	-1.031 m	≥ 0 → BRAKE	✗ → Brake activated
Final ped radius margin	3.555 m	≥ 0 m	✓ PASS
Outward main ay	2.2208 m/s ²	—	—
Outward counter ay	-0.8883 m/s ²	—	—
Return main ay	-2.2208 m/s ²	—	—
Return counter ay	0.8883 m/s ²	—	—
Outward counter scale α	0.400	[0.40, 0.85]	✓ (at clip)
Return counter scale α	0.400	[0.40, 0.85]	✓ (at clip)
Decision mode	BRAKE_WAIT_UNTIL_SAFE_THEN_OVERTAKE	—	✓



CARLA VALIDATION

Scenario 3



Thank You

